Renewi Contractor & Visitor Site Access

End-to-End Demo Product Requirements Document

A manager-ready demo blueprint based on the DigitalT3 discovery material and the supplied 11-step workflow image.

Purpose	Build a convincing end-to-end demo from scratch; not a production deployment.
Primary workflow	The supplied 11-step journey from Invite / request through Expiry / re-validation.
Pilot framing	Vlaardingen as the demo site, designed so the information architecture can scale to NL + BE.
Audience	Renewi / DigitalT3 manager and project stakeholders reviewing the working concept.
Source basis	DigitalT3 - Renewi - Vlaardingen Lighthouse discovery deck, prepared 28 Aug 2026.

IMPORTANT INTERPRETATION RULE

The discovery deck explicitly says the eleven-step journey is DigitalT3's reconstruction from the requirements sheet and a hypothesis to test in discovery. For this manager demo, that journey is nevertheless the controlling user-flow specification: build it exactly enough to make the end-to-end path visible and clickable, while keeping claims about the real Renewi as-is process clearly labelled as assumptions.

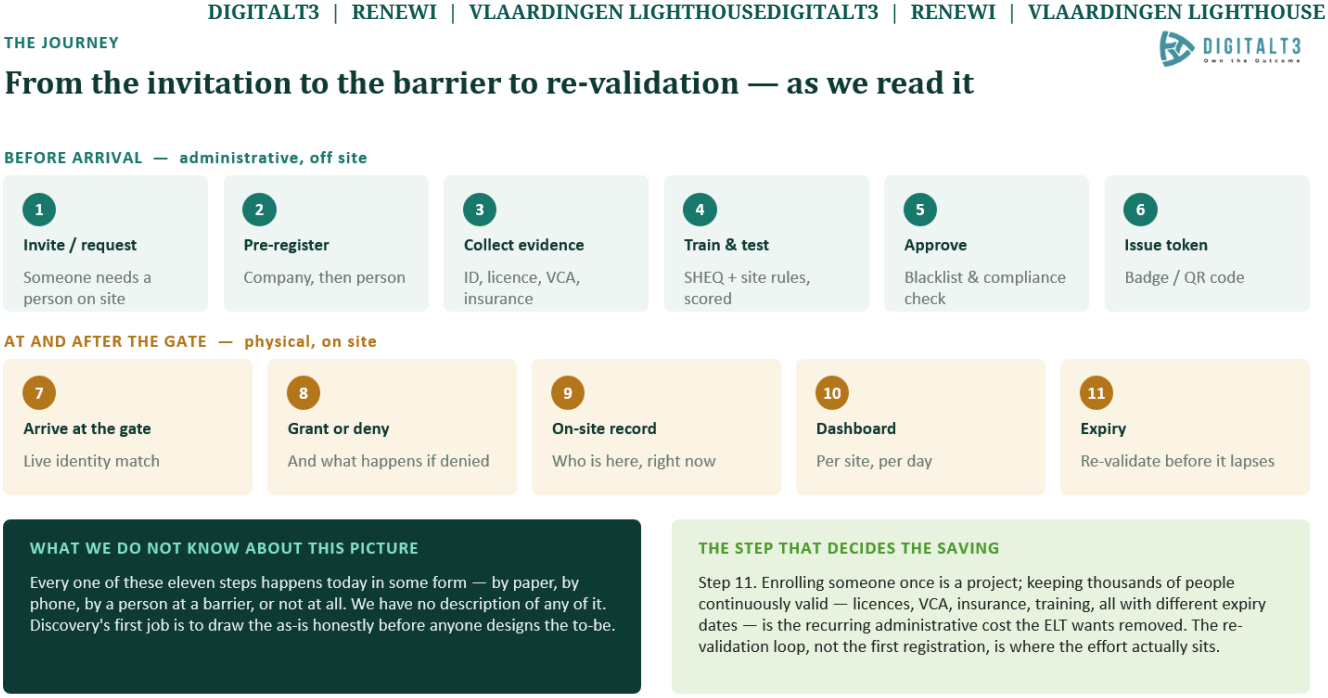


Figure 1. Primary demo workflow supplied by the project owner. Source: attached workflow image / DigitalT3 discovery material.

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BUILD PRINCIPLE

The product is not being judged on production infrastructure. It is being judged on whether a stakeholder can watch the entire journey work: invitation -> registration -> evidence -> training -> approval -> token -> gate decision -> on-site presence -> dashboard -> expiry / re-validation.

1. Product summary and source interpretation

The requested demo is a digital Contractor & Visitor Site Access platform for Renewi. The discovery material describes an ELT / board-driven goal: centralise compliance data and automate workflows so that only qualified and authorised people can access sites, while reducing administrative effort, improving safety and supporting regulatory compliance. The deck frames approximately 100 sites across the Netherlands and Belgium, with France and Portugal out of scope for now.

WHAT THE SOURCE DIRECTLY SUPPORTS

The deck defines six populations, a pre-registration layer, a population-specific check set, a daily per-site dashboard, and the 11-step journey shown in the supplied picture. It also states that the current-state process, systems, failure handling, exact compliance definitions and site realities are still open.

WHAT THIS PRD ADDS

This PRD turns the supplied discovery picture into a concrete demo specification: data entities, screens, APIs, decision rules, seeded scenarios, batch plan, requirements files, and a demo acceptance checklist. Those build details are implementation recommendations for the demo and are not assertions about Renewi's production estate.

Evidence from the discovery deck

Source area	What matters for the demo	Deck page
Executive summary	6 populations; NL + BE; ~100 sites; 1,500-2,500 people is a low-confidence estimate; Vlaardingen proposed as pilot.	pp. 3-4
Population scope	Renewi drivers, temporary/agency drivers, external drivers, maintenance contractors, permanent contractors, visitors.	p. 5
Check matrix	Corporate compliance, identity, driving licence, VCA, safety training, site rules, access token, blacklist.	p. 6
Journey	11 steps from Invite / request through Expiry / re-validation.	p. 7
Candidate MVP	Pre-register, capture evidence, train & test, approve, verify at gate, see who is on site.	p. 14
Risks	Privacy, estate variety, barrier adoption, duplication, unknown baseline, short-notice populations, language.	p. 15

2. Product vision, goals, scope and non-goals

2.1 Vision

Create one coherent system where a person can move from invitation to authorised access and then remain continuously valid through evidence expiry and re-validation. The manager should be able to see the workflow state at every step and demonstrate a grant and a denial without manual database manipulation.

2.2 Goals for the demo

- Demonstrate the full 11-step workflow with visible state transitions and audit history.
- Represent the six source populations in the domain model, while using a smaller demo dataset.
- Show evidence capture for ID / licence / VCA / insurance, with structured metadata such as expiry date and verification status.
- Show configurable training and site-rule tests with pass/fail results.
- Show approval, blacklist checking and QR/token issuance.
- Show gate evaluation with a simulated live identity match and explicit grant / deny reasons.
- Show a live on-site roster and per-site dashboard.
- Show expiry alerts and a re-validation loop that changes access status.
- Provide polished, production-style UI/UX and responsive behavior suitable for a manager presentation.

2.3 In scope

One pilot site (Vlaardingen) with multi-site-ready data structures; six populations; invitation and registration; evidence; training and tests; approval; tokens; gate evaluation; on-site presence; dashboard; audit log; expiry / re-validation; demo authentication and role-based navigation; synthetic demo data; API documentation; Docker-based local run.

2.4 Deliberately out of scope for this demo

- Physical barrier hardware integration or badge-reader procurement.
- Deep HR, agency, freight-exchange, LMS or enterprise identity integrations.
- Production biometric processing, retention, DPIA completion, or legal sign-off.
- France / Portugal rollout; Seveso-specific regime implementation.
- Authoring real SHEQ training content as an enterprise learning programme.
- Real customer / employee personal data. Use fictional demo records only.
- Production-grade multi-region HA, disaster recovery, SOC controls or 24/7 operations.

DEMO SUCCESS CRITERION

A manager should be able to start from an invited person, complete the prerequisites, issue a token, run the gate flow, see the person on the on-site roster, then advance time / trigger expiry and show that the same person must be re-validated before access is granted again.

3. Users, populations and role model

3.1 Source populations

Population	Definition	Visibility / notice	Demo treatment
Renewi driver	Own / permanent driver	Known, employed, on HR record	Full driver checks
Temporary driver	Supplied by employment agency	Known to agency, not necessarily to Renewi	Driver checks + external source path
External driver	Freight exchange	May be unknown until truck arrives	Short-notice path is critical
Maintenance contractor	Ad hoc specialist work	Short notice possible	Contractor checks
Permanent contractor	Fixed-term contract	Can be onboarded well in advance	Contractor checks + recurring validation
Visitor	Hosted visitor	Typically lightest check set	Identity + training/site rules + approval/token in demo

Source: discovery deck page 5. The different arrival-notice paths are an explicit DigitalT3 observation, not a confirmed current-state fact.

3.2 Demo roles

Role	Primary responsibility in demo
System Admin	Manage sites, users, configuration, demo data and feature toggles.
Compliance / SHEQ	Review evidence, manage training and test results, approve or reject compliance.
Site Manager	Review roster, site readiness, access decisions, exceptions and dashboard.
Gate Operator	Run scan / identity verification, view decision, record entry/exit and handle denial reason.
Requester / Host	Create invite, specify person/company/site/visit dates.
Contractor / Driver / Visitor	Complete registration, upload evidence, take training/tests, receive token.
Read-only Viewer	See dashboards, audit events and high-level status without mutation rights.

3.3 Navigation model

The left navigation should expose: Dashboard, People, Companies, Visits, Evidence, Training, Compliance / Approvals, Gate, On-site, Alerts / Re-validation, Audit, and Settings. Role-based permissions may hide or disable modules, but the underlying demo flow must remain navigable with seeded accounts.

4. The 11-step end-to-end workflow

PRIMARY WORKFLOW RULE

The supplied workflow image is the highest-priority build requirement. Build and test the system around these exact eleven states. The discovery deck describes the flow as a DigitalT3 reconstruction / hypothesis; the demo should therefore present it as the proposed workflow, not as a claim about today's Renewi process.

#	Step	Intent	Done when	Primary UI	State outcome
1	Invite / request	Requester creates a need for a person on site.	Invitation created	Invitation detail / create page	Invite -> Pending pre-registration
2	Pre-register	Company and person are registered.	Minimum person + company profile exists	Registration wizard	Pre-registered
3	Collect evidence	ID, driving licence, VCA, insurance as applicable.	Required evidence is uploaded and validated	Evidence workspace	Evidence complete / exceptions
4	Train & test	SHEQ + site rules; scored tests.	Required modules passed	Training & tests	Training complete
5	Approve	Blacklist + compliance checks.	All mandatory checks pass; approver signs off	Approval queue	Approved / rejected
6	Issue token	Badge / QR token is created.	Approved visit window exists	Token issue panel	Token active
7	Arrive at gate	Live identity match against registered identity.	Token scanned + identity check run	Gate kiosk / operator page	Gate evaluation pending
8	Grant or deny	Decision plus human-readable reason.	Rule engine returns result	Gate decision screen	Granted / Denied
9	On-site record	Who is on site right now.	Entry event recorded	On-site roster	Present
10	Dashboard	Per site, per day monitoring.	Events and states persisted	Dashboard	Current site picture
11	Expiry	Re-validate before validity lapses.	Any evidence / training / approval reaches expiry	Alerts + re-validation	Valid / Expiring / Expired

4.1 Required state machine

Person / visit lifecycle should be explicit and queryable, not inferred from UI. Recommended states: INVITED, PRE_REGISTERED, EVIDENCE_PENDING, TRAINING_PENDING, PENDING_APPROVAL, APPROVED, TOKEN_ISSUED, GATE_PENDING, GRANTED, PRESENT, EXITED, EXPIRING, EXPIRED, DENIED, BLACKLISTED, REVOKED. A person can return to a re-validation state without losing their historical audit trail.

4.2 Gate decision precedence

Priority	Condition	Failure behavior
1. Token	Token exists, active, correct site and current time within visit window.	Deny: invalid / expired / wrong site token
2. Person status	Person is approved, not revoked / blacklisted.	Deny: not authorised / blacklisted
3. Evidence	Mandatory evidence is present, verified and unexpired.	Deny: missing / invalid / expired evidence
4. Training	Required modules and tests are complete and valid for site.	Deny: training incomplete / expired
5. Identity	Presented identity matches registered identity in demo verifier.	Deny: identity mismatch
6. Final decision	All required checks pass.	Grant entry + create on-site record

5. Functional requirements

ID	Area	Requirement	Priority
FR-01	Authentication	Provide seeded demo accounts with role-based access; no external identity provider required.	High
FR-02	Invite	Create invitation with requester, company, person (optional), site, visit window, population type and purpose.	High
FR-03	Pre-registration	Create or link company and person records; prevent duplicate person records in demo by normalized email / phone / external ref.	High
FR-04	Company compliance	Record corporate compliance status, insurance, approval state and evidence metadata.	Medium
FR-05	Evidence intake	Upload synthetic ID, driving licence, VCA, insurance docs; classify document type, issue date, expiry date, verification result.	High
FR-06	Evidence extraction	Expose extracted fields in editable review UI; allow approve / reject / request re-upload.	High
FR-07	Training catalogue	Create site-specific and global training modules with validity period and language metadata.	High
FR-08	Training completion	Mark modules complete; store test attempts, score, pass/fail and completion time.	High

ID	Area	Requirement	Priority
FR-09	Approval	Run compliance summary, blacklist check and approver action with reason.	High
FR-10	Blacklist	Maintain person and company blacklist entries with active/inactive state and reason.	High
FR-11	Token issuance	Issue signed / opaque QR access token tied to person + site + visit window.	High
FR-12	Gate scan	Accept QR token input or camera-scan simulation; retrieve visit and person context.	High
FR-13	Identity verification	Provide a demo verifier adapter that compares presented identity to registered identity and returns match / mismatch + confidence.	High
FR-14	Gate decision	Evaluate all mandatory checks and show Grant / Deny plus reason codes and next action.	High
FR-15	Gate event	Record arrival, entry, exit, denied attempt, override and operator identity.	High
FR-16	On-site roster	Show who is currently present, entry time, population, company, token status and last verification.	High
FR-17	Dashboard	Show site/day metrics, current presence, pending approvals, expiring evidence and denial trend.	High
FR-18	Expiry engine	Calculate upcoming / expired evidence, training, approval and token dates.	High
FR-19	Re-validation	Create re-validation tasks; approve updated evidence / training and restore gate eligibility.	High
FR-20	Audit	Append immutable-ish demo audit events for important actions.	High
FR-21	Search	Global search across person, company, visit, token and document number / reference.	Medium
FR-22	Demo controls	Provide a safe demo-only time shift / scenario reset mechanism behind admin access.	High

6. Business rules and compliance matrix

6.1 Source check matrix carried into the demo

Check	Applies to (source)	Demo interpretation	What the UI stores
Corporate compliance	Drivers, Contractors	Optional / not applicable	Company supplier standing, legal / insurance status
Personal identification	Drivers, Contractors, Visitors	All	Registered identity vs presented identity
Licence to drive	Drivers	Not applicable	C / CE / Code 95 and in date
VCA	Contractors	Not applicable	Safety certification for contractor personnel
Safety training	Drivers, Contractors, Visitors	All	SHEQ-related training + scored test
Site rules	Drivers, Contractors, Visitors	All	Site-specific rules + scored test
On-site access approved	Drivers, Contractors, Visitors	All	Badge / QR / equivalent token
Blacklist	Drivers, Contractors, Visitors	All	Rejected people and/or companies

The discovery deck says blank cells in the population/check matrix are DigitalT3 inference. The demo should therefore store applicability as configuration, not hard-code it in presentation components.

6.2 Rule configuration

Each site should have a compliance profile containing required checks by population. Each requirement has: required flag, evidence type / training module, verification method, validity window, renewal lead time, language, and blocking severity. This permits a single code path for drivers, contractors and visitors while the site profile controls the differences.

6.3 Demo rule examples

Demo scenario	Required checks	Primary denial triggers	Decision
Driver at Vlaardingen	ID + licence + safety training + site rules + blacklist + approval + active token	Block on licence expiry; block on failed training; block on blacklist.	GRANT only if all mandatory checks pass.
Contractor at Vlaardingen	ID + VCA + safety training + site rules + blacklist + approval + active token	Block on VCA expiry or missing training.	GRANT only if all mandatory checks pass.
Visitor at Vlaardingen	ID + safety training + site rules + blacklist + approval + active token	Do not require driving licence / VCA in default demo profile.	GRANT only if all configured visitor checks pass.

7. UI/UX and design system requirements

MANAGER PRESENTATION STANDARD

The UI must look production-grade even though the backend is a demo. Use a restrained visual system: dark teal as the primary brand tone, warm neutral / amber accents for physical gate states, light surfaces, clear status badges, generous spacing, high-quality tables and strong empty / loading / error states.

7.1 Global UX rules

- Desktop-first responsive layout for manager presentations, with tablet-friendly breakpoints.
- Persistent left navigation, compact top bar, breadcrumb / page title, and contextual actions.
- Every workflow item has a visible status, last updated time and owner.
- Use progressive disclosure: summary cards first, detailed evidence and audit history one level deeper.
- Avoid long forms: use multi-step wizards with autosave and validation indicators.
- Use human-readable reason codes on every denial, e.g. “Driving licence expired 14 days ago”.
- Use clear “what happens next” actions so the demo can be driven without explanation.

7.2 Required screens

Screen	Purpose	Key interaction
Login	Role selection through seeded credentials; branded visual.	Seeded demo accounts
Dashboard	Site overview, present count, compliance state, alerts, denials, upcoming expiry.	Site + date filter
People	Searchable directory with status, population, company and compliance score.	Filter by population / status
Person detail	Profile, company, visits, evidence, training, approvals, tokens, audit.	Timeline is central
Invitation	Create request, assign site, dates, type, host/requester.	Wizard
Evidence	Document cards, extraction fields, expiry countdown, review action.	Drag-drop + sample docs
Training	Modules, progress, test score, pass/fail, renewal date.	Checkbox completion + score simulation
Approval queue	List of pending people, checklist, blockers, approve / reject.	One-click decision with reason
Token	Issue / revoke QR, show token validity and print/share demo card.	QR visual
Gate	Camera / scan simulation, identity match, decision banner, event actions.	Designed like a kiosk
On-site	Live roster, entry/exit, duration, exception flags.	Auto-refresh demo
Alerts / Re-validation	Expiring items grouped by urgency; launch re-validation.	30/14/7 day bands
Audit	Chronological event stream with actor, action, entity and reason.	Export as CSV optional
Settings	Site profile, rules, training modules, role mapping, demo controls.	Admin only

7.3 Gate UX

The gate screen is the most important single screen after the workflow. It should support three visible stages: Scan -> Verify -> Decision. The final decision should dominate the screen, with a short reason list, operator identity, timestamp, site and next action. “Denied” must not be a dead end: present the appropriate action such as “Resolve expired licence”, “Send to registration”, “Contact site manager”, or “Manual review”.

8. Technical architecture and recommended tech stack

8.1 Recommended stack

Layer	Recommendation	Why it fits the demo
Web app	Next.js + TypeScript	App Router; server/client components where useful; route-based navigation.
UI	Tailwind CSS + shadcn/ui	Professional, accessible components; simple design tokens.
Forms / validation	React Hook Form + Zod	Consistent client-side form schemas and UX.
Data fetching	TanStack Query	Caching, mutations, loading and error state handling.
API	FastAPI + Pydantic	Python backend with OpenAPI documentation.
ORM / migrations	SQLAlchemy + Alembic	Explicit relational model and repeatable migrations.
Database	PostgreSQL	Relational source of truth for people, companies, visits, evidence, training and events.
Storage	Local filesystem adapter + S3-compatible abstraction	Fast demo setup with migration path to object storage.
OCR / document intelligence	Provider adapter; mock by default, optional Anthropic vision adapter	Demo works without credentials; provider can be enabled later.
Identity matching	Demo adapter + optional CV provider interface	Avoid real biometric dependency for the demo.
QR	Python qrcode / browser QR library	Generate visible QR tokens tied to visit authorisation.
Background work	APScheduler / lightweight scheduled worker	Expiry scanning and alert generation; manual trigger available for demos.
Tests	Pytest + Playwright	API/service tests plus end-to-end browser scenario.
Packaging	Docker Compose	One-command local boot for web, API and database.
API docs	OpenAPI / Swagger via FastAPI	Makes the demo architecture legible to technical reviewers.

8.2 Logical architecture

Browser -> Next.js web app -> FastAPI API -> domain services -> PostgreSQL / file storage. The API exposes provider adapters for OCR, identity verification, notification and token delivery so the system can run in

deterministic mock mode. The domain layer owns compliance decisions; UI components must never re-implement gate rules.

ARCHITECTURE CONSTRAINT

Keep all external capabilities behind interfaces. The demo must boot and complete the 11-step journey with zero third-party credentials. Environment variables may switch provider adapters on, but missing keys must fall back to deterministic demo providers with clear “DEMO MODE” labeling.

8.3 Suggested repository structure

- apps/web/ - Next.js application
- services/api/ - FastAPI application and domain services
- services/worker/ - expiry / notification jobs (can be merged into API for first demo)
- database/migrations/ - Alembic migrations
- database/seed/ - deterministic demo seed data
- storage/demo/ - synthetic documents / images / QR assets
- requirements/ - build requirements, credentials, environment, acceptance criteria
- docs/ - architecture and operator notes
- tests/e2e/ - Playwright end-to-end workflow
- docker-compose.yml - local environment

9. Data model

Entity	Key fields	Purpose
Person	id, name, email, phone, DOB optional, population_type, status, registered_photo_ref, company_id	Core identity record
Company	id, name, type, supplier_status, blacklist_status, contact	Company-level compliance
Site	id, name, country, timezone, active, gate_mode	Site context
Visit	id, person_id, site_id, visit_type, purpose, start_at, end_at, status, host_id	A concrete site visit
Invitation	id, requester_id, person_id, company_id, site_id, requested_at, status	Step 1 record
Evidence	id, person_id, type, file_ref, extracted_fields_json, verified, issued_at, expires_at, status	Licence / VCA / ID / insurance
TrainingModule	id, site_id nullable, code, title, language, validity_days, passing_score, active	Training catalogue
TrainingResult	id, person_id, module_id, score, passed, completed_at, expires_at, attempt_no	Step 4 evidence
Approval	id, visit_id/person_id, decision, approver_id, decided_at, reasons	Step 5 decision
BlacklistEntry	id, person_id nullable, company_id nullable, active, reason, starts_at, ends_at	Blocking rule
AccessToken	id, visit_id, token_hash, display_code, qr_payload, issued_at, expires_at, status	Step 6
GateEvent	id, visit_id, token_id, event_type,	Steps 7-9 audit / presence

Entity	Key fields	Purpose
	identity_match, decision, reasons_json, operator_id, occurred_at	
RevalidationTask	id, person_id, evidence_id/training_id nullable, due_at, status, completed_at	Step 11
AuditEvent	id, actor_id, action, entity_type, entity_id, payload_json, occurred_at	Full traceability
DemoScenario	id, name, description, seed_state, enabled	Manager demo reset / scenario control

9.1 Data integrity rules

- Never delete compliance evidence or gate events in normal UI flows; use status / supersede semantics.
- All dates are timezone-aware. Site timezone is the authority for gate decisions and day dashboards.
- Token payload should not contain raw personal data; store a reference and resolve server-side.
- Every gate decision stores reasons at the time of decision so later rule changes do not rewrite history.
- A person may have multiple visits and tokens, but only one current master person record per identity reference in the demo.

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10. API / service contracts

Method	Endpoint	Contract / intent
POST	/auth/login	Authenticate demo account; return session / JWT.
GET	/sites	List sites.
GET	/people?status=&population=	Search people.
POST	/invitations	Create invitation / request.
POST	/registrations/{person_id}	Complete pre-registration.
POST	/evidence	Upload or attach demo evidence metadata.
POST	/evidence/{id}/extract	Run provider adapter and return structured fields.
PATCH	/evidence/{id}/verify	Approve / reject evidence.
GET	/training/modules	List modules for site/population.
POST	/training/results	Record completion / test score.
POST	/approvals/evaluate	Calculate current compliance summary.
POST	/approvals/{id}/decision	Approve / reject with reasons.
POST	/access-tokens	Issue access token.
POST	/access-tokens/{id}/revoke	Revoke token.
POST	/gate/evaluate	Run token + compliance + identity decision.
POST	/gate/events	Record entry / exit / denial / override.
GET	/on-site?site_id=	Return current presence.
GET	/dashboard?site_id=&date=	Return KPI cards, activity, roster and alerts.
GET	/revalidation/tasks	Return expiring / expired tasks.
POST	/revalidation/run	Run expiry scan for demo.
POST	/revalidation/{id}/complete	Complete a re-validation task.
GET	/audit	Query audit events.
POST	/demo/scenarios/{name}/reset	Reset deterministic demo scenario.

10.1 Service boundaries

- ComplianceService: resolves the configured rule set and returns a typed decision with reasons.
- EvidenceService: handles upload metadata, extraction adapter and verification lifecycle.
- TrainingService: manages modules, attempts, scores and expiry.
- GateService: orchestrates token, status, evidence, training and identity checks.
- PresenceService: creates / closes on-site records from gate events.
- RevalidationService: detects upcoming expiry and creates tasks / alerts.
- AuditService: appends domain events for traceability.

11. Demo scenarios and acceptance criteria

11.1 Mandatory demo personas

Persona	Population	Scenario	Expected outcome
Alex van Dijk	External driver	Happy path	All evidence valid; training passed; approved; token active; gate GRANT.
Sofia Janssen	Contractor	Expired evidence	VCA expired; gate DENY with clear action.
Milan de Boer	Visitor	Training incomplete	Site rules test not passed; approval blocked or gate DENY.
Noah Peters	Driver	Blacklist	Active blacklist entry; gate DENY even with valid token.
Emma Visser	Contractor	Identity mismatch	Token valid but presented identity does not match; gate DENY.
Lucas Smit	External driver	Revalidation	Evidence expires; task created; access becomes blocked until renewed.

11.2 Manager demo script

1. Open Dashboard for Vlaardingen and show current site KPIs.
2. Open Alex van Dijk -> show profile, company, evidence, training and approval timeline.
3. Open Access Token -> display QR and visit window.
4. Switch to Gate -> scan / paste token -> run identity match -> show green GRANT.
5. Record entry -> open On-site -> show Alex as PRESENT.
6. Open Dashboard -> show roster and recent gate event.
7. Switch to Sofia -> show expired VCA -> run Gate -> show DENY with reason and action.
8. Switch to Noah -> show blacklist -> show DENY without needing a new token.
9. Open Re-validation -> run expiry scan -> show Lucas in expiring / expired queue.
10. Complete Lucas re-validation -> re-run Gate -> show access restored.
11. Open Audit -> show the complete trace from invite through gate and re-validation.

11.3 Acceptance criteria

ID	Acceptance criterion
AC-01	A new invitation can be created and appears as Step 1 / Invited.
AC-02	The invitation can be converted into a person + company registration and shown as Step 2.
AC-03	At least four evidence types can be attached and expiry dates are visible.
AC-04	Training and site-rule tests can be marked complete with a score and pass/fail state.
AC-05	Approval shows a checklist and reasons; blacklist blocks approval.

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ID	Acceptance criterion
AC-06	An approved visit can generate a QR token with validity window.
AC-07	Gate evaluates the token and compliance server-side; UI only renders the decision.
AC-08	At least one happy path shows GRANT and creates a presence record.
AC-09	At least four denial paths work: expired evidence, missing training, blacklist, identity mismatch.
AC-10	Dashboard shows current on-site people and gate activity for the site/day.
AC-11	Expiry scan creates re-validation tasks and changes access eligibility when a blocking item expires.
AC-12	Completing re-validation restores eligibility and is visible in Audit.
AC-13	The whole flow works from seeded data without any external API key.
AC-14	UI has loading, success, error, empty and permission states for all major pages.
AC-15	Docker Compose boot + database migration + seed can be repeated from a clean environment.

12. Batch implementation plan

EXECUTION RULE FOR THE BUILDER

Build in batches, but do not leave a batch half-finished. At the end of each batch: run the specified validation, fix failures, update the requirements / status files, and only then begin the next batch. The manager demo is successful only when the final batch exercises the entire 11-step workflow.

Batch	Focus	Deliverables	Exit gate
Batch 0	Foundation	Monorepo structure, Docker Compose, PostgreSQL, FastAPI health, Next.js shell, environment templates, linting, formatting, seed / reset framework.	Clean boot; /health works; UI loads; DB migrates.
Batch 1	Domain + auth	Database schema, migrations, seed data, roles, login, navigation, sites, people, companies, visits, audit base.	CRUD / read flows work; seeded users can navigate.
Batch 2	Steps 1-3	Invitations, pre-registration wizard, evidence intake, document cards, extraction adapter and expiry metadata.	Create person from invite and attach evidence end-to-end.
Batch 3	Steps 4-6	Training modules, test engine, approval queue, blacklist, compliance summary, QR token issuance / revoke.	Approved person gets usable token; blocked person cannot.
Batch 4	Steps 7-8	Gate kiosk, QR scan simulation, demo identity match, compliance gate service, grant / deny reasoning, operator actions.	Happy path grant + denial scenarios all work server-side.
Batch 5	Steps 9-10	On-site roster, entry / exit state, dashboard, filters, daily site metrics, recent events.	Gate event changes roster and dashboard immediately.
Batch 6	Step 11	Expiry engine, alert bands, re-validation queue, renewal actions, access re-evaluation.	Expired item blocks gate until revalidated; audit persists.
Batch 7	Production-style polish	Design system cleanup, responsive layout, empty/loading/error states, accessibility, animations, table polish, demo mode labeling.	Manager-ready visual finish.
Batch 8	QA + demo packaging	Playwright full workflow, API tests, scenario reset, README, requirements docs, screenshots, final smoke test.	One-command demo and reproducible manager walkthrough.

12.1 Validation command set

The final repository should provide explicit commands (or Makefile / PowerShell equivalents) for: install, dev, db migrate, db seed, db reset, lint, typecheck, unit tests, API smoke test, Playwright E2E, Docker boot, and demo scenario reset. The exact commands may vary by implementation, but they must be documented and executable by a fresh user.

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13. Requirements and credentials specification

CREDENTIAL PHILOSOPHY

No credential should block the build. Every external integration is optional and must have a deterministic demo adapter. Create the requirements documentation before or alongside Batch 0, and keep it updated after every implementation batch.

13.1 Required project files

File	Purpose	Rule
requirements/README.md	Single source of truth for setup requirements, dependencies and implementation notes.	Must be created in Batch 0.
requirements/CREDENTIALS.md	Provider, variable name, where to obtain it, why it is used, whether optional, safe placeholder.	No real secrets committed.
requirements/ENVIRONMENT.md	Local / demo / integration environments and required configuration.	Include .env.example references.
requirements/DEMO_DATA.md	Synthetic personas, documents, site, training and scenario definitions.	Must map to manager demo script.
requirements/INTEGRATIONS.md	Adapters for OCR, storage, notifications, identity matching and future HR/LMS feeds.	Mock-first.
requirements/ACCEPTANCE_CRITERIA.md	Machine-readable / checkable copy of the acceptance criteria in this PRD.	Update each batch.
requirements/OPEN_QUESTIONS.md	Source uncertainties and decisions that would be required for a production build.	Keep demo assumptions explicit.

13.2 Credentials inventory

Variable family	Provider	Purpose	Demo rule
DATABASE_URL	PostgreSQL	Required for DB access	Demo provides local container value.
APP_SECRET_KEY	Application	Session / JWT signing	Generate locally; not external.
STORAGE_*	Object storage (optional)	Upload adapter	Local filesystem is default.
ANTHROPIC_API_KEY	OCR / document vision (optional)	Real provider adapter for document extraction	Demo must work without it; label as optional.
IDENTITY_PROVIDER_*	Identity / CV (optional)	Optional future gate verification provider	Demo adapter by default.
SMTP_*	Email (optional)	Invitation / alert delivery	Use in-app notification panel or local mail catcher for demo.

The specific use of Anthropic is an optional implementation recommendation, not a statement that Renewi has selected it. The source deck only requires document intelligence and identity / live match capabilities.

13.3 Demo document set

- Synthetic ID card image for identity record.
- Synthetic Dutch-style driving-licence mockup for demo OCR / field extraction; clearly fictional and not a real credential.

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- Synthetic VCA certificate.
- Synthetic insurance certificate.
- Synthetic training completion / site rules material.
- QR access token graphics generated by the application.

14. Security, privacy and demo safeguards

The discovery material explicitly flags identity documents and live identity matching as personal-data / potentially biometric concerns and says the legal basis, retention and DPIA route remain open. This demo therefore must not pretend to be production-compliant.

Control	Demo implementation	Priority
Synthetic data only	No real employee / driver identity documents or personal data in the repository.	Mandatory
Secrets	Use .env.example; never commit API keys or passwords.	Mandatory
RBAC	Enforce permissions in the API, not only the UI.	Mandatory
Audit trail	Log sensitive actions such as approval, blacklist, token issue, gate decision and re-validation.	Mandatory
Token safety	Use non-guessable references; do not expose personal data inside QR payload.	Mandatory
Uploads	Restrict file type / size; keep demo storage outside public static path.	Recommended
Retention	Use a simple documented demo retention policy and label it “demo only”.	Mandatory
Biometrics	Do not require production-grade face recognition; use a deterministic simulated verifier.	Mandatory

DO NOT CROSS THE DEMO BOUNDARY

The product can demonstrate the control flow of identity matching without collecting or processing real biometric data. The visual “live match” should be a simulation or synthetic comparison layer, clearly marked as DEMO MODE.

15. Risks, assumptions, open questions and future scale

15.1 Risks directly identified in the source

Risk	Source concern	Demo design response
Duplication	Existing contractor / Seveso regime or vendor may already cover part of the scope.	Demo remains modular; do not hard-code external-system assumptions.
Privacy	Identity capture / live match may require DPIA and legal decisions.	Synthetic data only; adapter boundary.
Estate variety	~100 sites may have different gate hardware, network and staffing.	Site config + gate mode in data model.
Barrier adoption	A denial can stop a truck / person at the gate.	Fast decision UI + explicit fallback path.
No baseline	Current-state effort and volumes are not yet measured.	Demo metrics are synthetic and labelled.
Uncontrolled populations	Agency / freight-exchange drivers may be outside Renewi systems.	Manual enrolment is first-class in demo.
Language	Drivers may span NL, BE and other languages.	Training modules carry language metadata.

15.2 Key source questions kept open for production discovery

- Is Vlaardingen confirmed as the pilot site?
- Does an existing contractor-compliance or site-access capability already exist, especially at Seveso sites?
- What is the authoritative source for people and companies?
- What exactly happens at the gate today, including offline / no-network cases?
- Who owns each compliance decision and who can overturn a blacklist?
- Where do SHEQ training and tests live today?
- What volumes exist per population / site / day and what is the repeat-visit ratio?
- Is a DPIA required for identity capture and live matching, and what is the legal basis / retention model?
- What enterprise IT standards / hosting constraints apply?

15.3 Scale path after the demo

The data model should scale by population, site, integration and country. First add more population-specific journeys, then multi-site readiness, then HR / agency feeds, then country-specific legal rules. France and Portugal remain outside this scope, consistent with the source material.

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16. Definition of done

16.1 Product DoD

- All 11 workflow steps have a dedicated UI state and backend state transition.
- The happy path and at least four failure paths are reproducible with seeded demo data.
- The gate decision is a backend/domain decision, not front-end-only logic.
- On-site roster and dashboard update from recorded gate events.
- Expiry and re-validation change gate eligibility and are auditable.
- The application can run without third-party credentials.
- All major screens include loading, empty, error and permission states.

16.2 Engineering DoD

- Database migration is repeatable from empty database.
- Seed script is deterministic and resettable.
- API schemas are typed and documented in OpenAPI.
- Unit tests cover core compliance decision paths.
- One Playwright E2E test covers the entire happy path from invite to re-validation.
- Repository has setup documentation, environment template and requirements folder.
- No secrets or real personal documents are committed.

16.3 Manager demo DoD

READY TO PRESENT WHEN…

A manager can sit beside the operator and follow the eleven numbered steps on screen, with no hidden admin database actions required to make the story work. The demo must show both a successful person and a refused person, and must finish by showing that expired evidence creates ongoing administrative work that re-validation resolves.

Appendix A - Source-to-PRD traceability

Deck page(s)	Source content used	PRD treatment
Page 3	Overall ask, six populations, NL/BE, ~100 sites, pilot framing	Sections 1-3
Page 5	Six populations and different arrival notice	Section 3
Page 6	Compliance checks by population	Section 6
Page 7	11-step journey	Section 4; primary acceptance focus
Pages 8-10	Open process, ownership, systems, law, site questions	Sections 14-15
Pages 11-13	Discovery workstreams and deliverables	Section 15; production discovery boundary
Page 14	Candidate MVP slice	Sections 2, 11, 12
Page 15	Risks and principles	Sections 7, 14, 15
Page 18	What is stated vs inferred	Used to label hypotheses / assumptions throughout

This PRD intentionally preserves the distinction the deck makes between statements in the requirements sheet and DigitalT3 inference. The build specification is more concrete because a working demo requires explicit rules, but those concrete rules should be described to stakeholders as demo behavior unless validated in a later discovery.

Appendix B - Builder prompt packaging

The companion build prompt supplied in chat should be treated as the execution instruction set for an AI coding agent. It repeats the workflow priority, architecture, requirements-folder obligations, batch plan, demo scenarios and definition of done from this PRD so the implementation remains aligned to the manager-facing story.